Understanding Why an NFL MVP is picked

2026-11-07

Abstract

Within our project, we analyze the five most recent NFL MVPs in order to find what kind of correlation QB statistics have with winning the NFL MVP. Currently we've seen a commonality that no matter what the MVP is leading their team to winning record, i.e above ".500". Another factor noted within our data is that the TD-INT ratio is also a factor, this being how many times a Quarterback has a touchdown pass per interception. Another is that the QB who won the award was usually at least top 10 when it came to most statistics like pass yards, Pass touchdowns, and other offensive metrics.

Introduction

The central question we have is: what decides who wins the MVP at the end of the regular season? From sports fans who may just be casuals and don't have the time to watch every game to die-hard fans who study film, we believe this analysis can help them decide if their QB has a chance at winning the MVP, and maybe anybody else interested in the MVP race. Of course, just looking at a couple of years of data doesn't do much until we've turned it into useful graphs; our primary objective was to classify what stood out about those individuals who had won MVP and what put them above others, and what statistics we considered valuable.

Summary

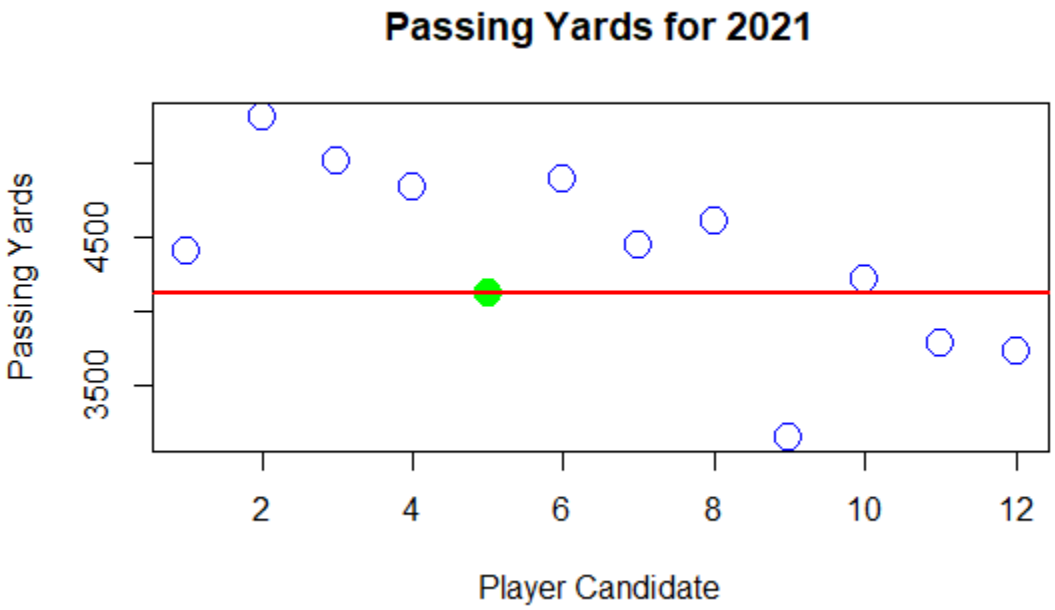
Being top 10 in most, if not all, QB statistics Having a high TD-INT ratio, i.e don't turn the ball over *QB must be on a team with a winning record, i.e.,>.650

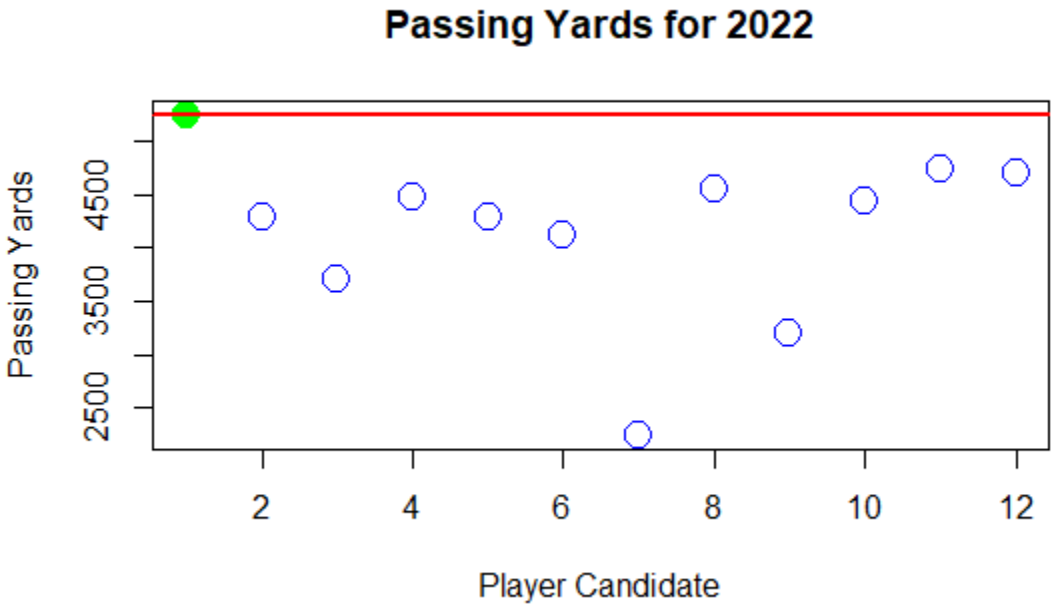
Data

The available data to us is from decades of MVP races; we specifically gathered our data from pro-football-reference.com, which has a wide and detailed history of NFL statistics. We came to a total of 60 observations with them spanning 6 significant variables to help determine our answer. The data cleaning process was not difficult and was made easy by the website because it has an option to export these statistics and allows you to

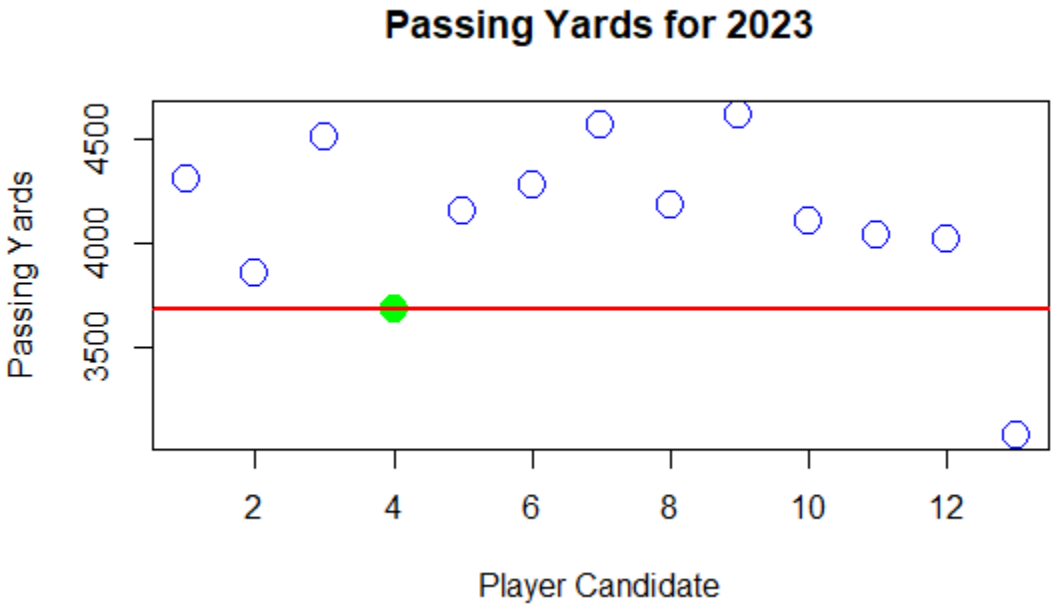
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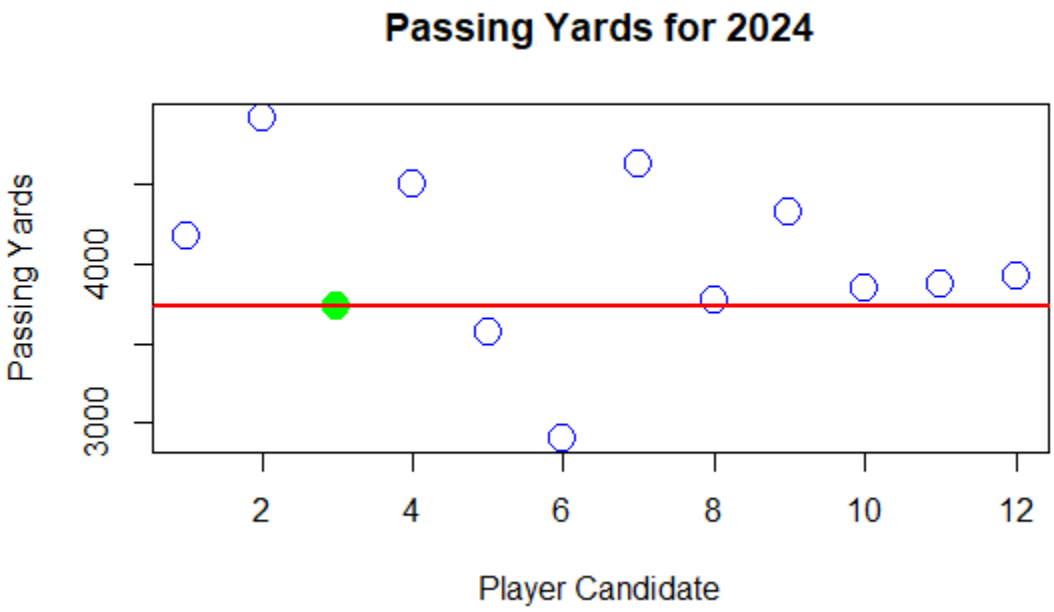
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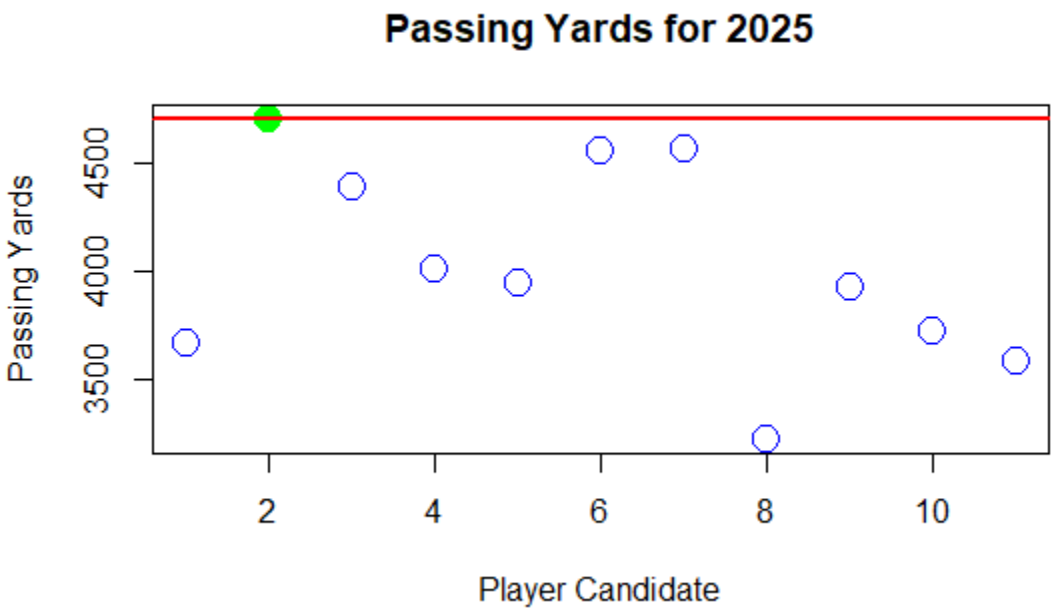


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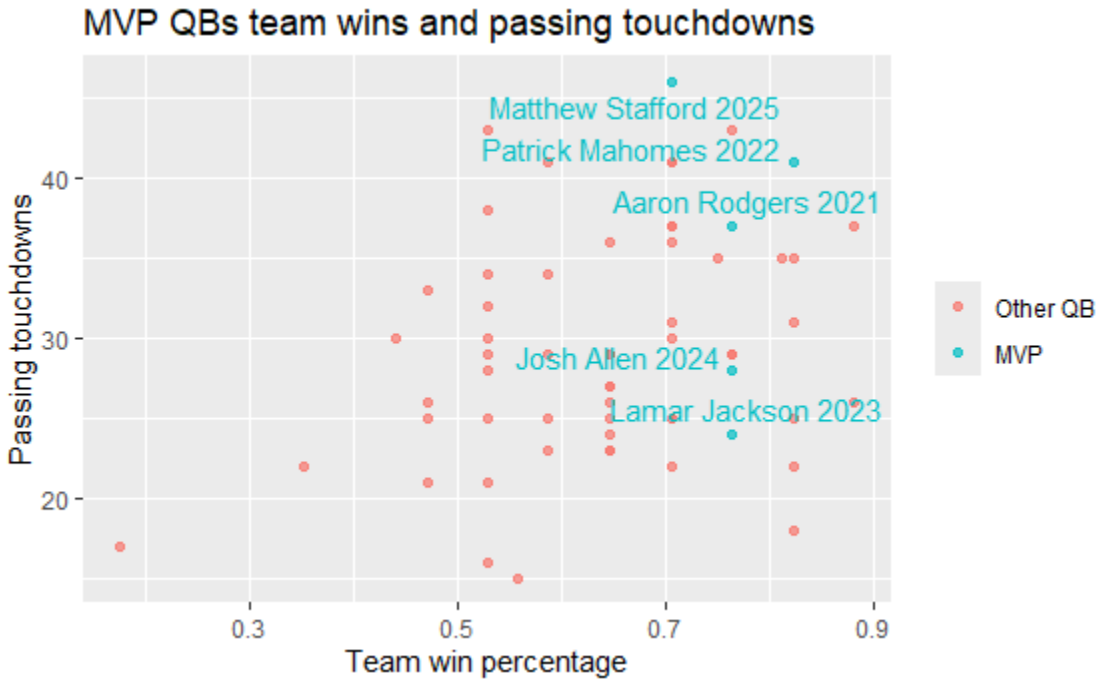




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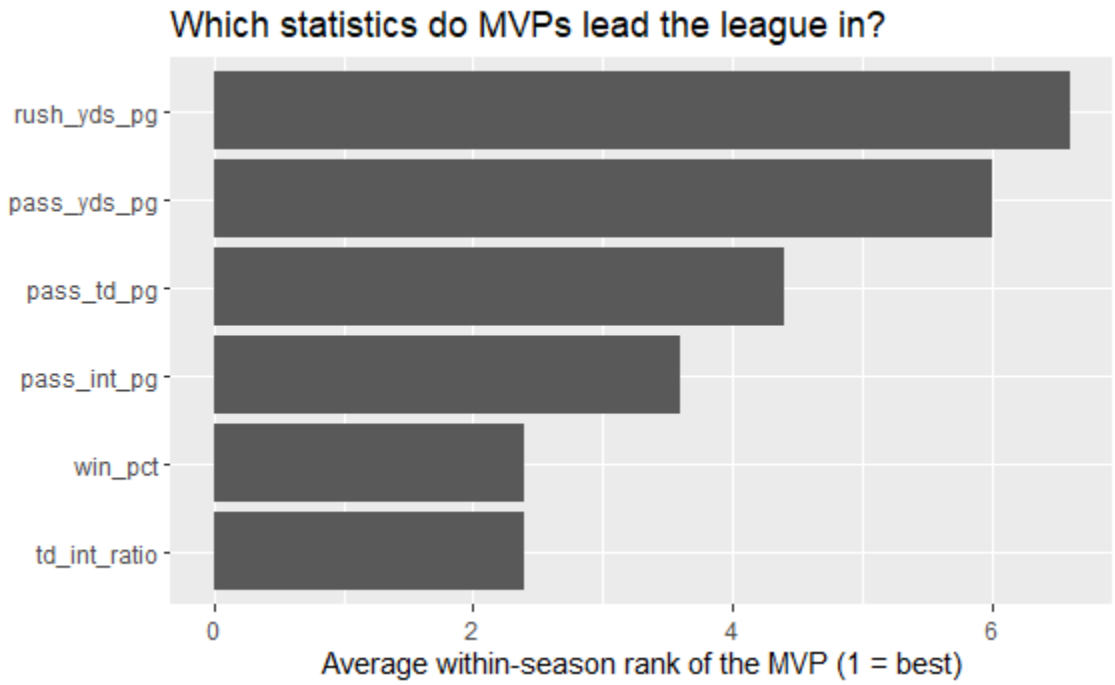


The Graphs above are merely to illustrate that the MVP does not need to have 1st in all statistics, but instead, when we take a look at the graphs below, we can see commonalities among the MVPs



We can see unanimously among the QBs that they lead their teams to be at least a .700+ record in the regular season, showing that MVPs are always on teams with a majority of their games.

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Analysis

Another key comparison made here was the difference in which statistics the MVP is most exclusive, specifically when looking at the win percentage and TD-int ratio, which are the most significant. The reason behind this is that win percentage demonstrates that the QB themselves can lead their team to a winning record, which is important when it comes time for the playoffs and their seeding. The other statistic, being the ratio is important to show case how impact they are to not just being an offensive necessity but also no being a liability in turning the ball over. We expect to see similar results if we were to expand our range on years for the MVP, and our comparison methods could be improved by using a regression model to get a definitive answer on what statistics are most impactful, or also doing doing an expansion on the variables to see what other QB statistics may be significant.

Credit

All: Helped draft topic, find data, and identify necessary information for project

Marcus Poon: README.md file 00_requirements.R file 01_funct_DataCleaning.R file 11_DataCleaning.Rmd file 12_DataSaving.Rdata file

Janelle Valadez 31_DataVisualization.Rmd <-first 30 graphs doen by janelle, last 3 and numerical comparison done by Marucs

German Gonzalez: FinalReport.Rmd

GitHub link: https://github.com/janellev06/Final_STAT107

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